

International Islamic University Chittagong
Department Of Computer Science and Engineering
Differential Equation
Ordinary Differential Equation
Group- A

1. Define differential equation. Classify the differential equation with example.

Formation of Differential Equation

1. Form the differential equation whose solution is given by
 - i. $y = A \cos (ax - b)$ (Au-23)
 - ii. $xy = pe^x + qe^{-x}$
 - iii. $y = A \cos px + B \sin px$
 - iv. $y^2 = Ax^2 + Bx + C$
 - v. $y = ae^{3x} + be^x$
2. Define Order and Degree of a differential equation with example. (Au-23)
3. Determine the order and degree of the following D.E.

First Order First Degree Differential Equation

Variables Separable

1. $x(y - 3) dy = 4y dx$.
2. $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$
3. $\sin y \frac{dy}{dx} = x^2$
4. $(1+y) \frac{dy}{dx} = (1 - x^2)$
5. $y \frac{dy}{dx} = x+1$
6. $\frac{dy}{dx} - y \tan x = 0$
7. $\frac{dy}{dx} = xy^3(1+x^2)^{-\frac{1}{2}}, y(0) = 1$

Reducible Separable Variable

1. $\frac{dy}{dx} = (4x+y+1)^2$

Homogeneous Differential Equation

1. $y^2 + x^2 \frac{dy}{dx} = xy \frac{dy}{dx}$
2. $\frac{dy}{dx} = \frac{-(x^2+y^2)}{2xy}$
3. $\left(x \cos \frac{y}{x} + y \sin \frac{y}{x}\right) y = \left(y \sin \frac{y}{x} - x \cos \frac{y}{x}\right) x \frac{dy}{dx}$ (Au-23)

4. $\frac{dy}{dx} = \frac{x^2 + 3y^2}{2xy}$
5. $\frac{dy}{dx} = \frac{x + 3y}{x - y}$

Reducible Homogeneous Differential Equation

1. $(2x + y + 3) \frac{dy}{dx} = x + 2y + 3$
2. $(2x - 2y + 5) \frac{dy}{dx} = x - y + 3$ (Au-23)
3. $(3x - 7y - 3) \frac{dy}{dx} = 3y - 7x + 7$
4. $(2x + y + 3) \frac{dy}{dx} = x + 2y + 3$

Linear Differential Equation

1. $(1 - x^2) \frac{dy}{dx} - xy = 1$
2. $x \frac{dy}{dx} + 2y = x^2 \log x$.
3. $y \log y \, dx + (x - \log y) \, dy = 0$
4. $\frac{dy}{dx} - \frac{1}{x} y = x^2$
5. $\frac{dy}{dx} + xy = x$
6. $\frac{dy}{dx} + y = \frac{1}{1 + e^{2x}}$
7. $\frac{dy}{dx} + \cot x \cdot y = 2x \operatorname{cosec} x$
8. $x \frac{dy}{dx} - y = x^2 \sin x$

Bernoulli's Differential Equation

1. $\frac{dy}{dx} = x^3 y^3 - xy$ (Au-23)
2. $\frac{dy}{dx} + xy = xy^4$
3. $x \frac{dy}{dx} + y = x^4 y^3$
4. $xy^2 \frac{dy}{dx} + y^3 = x \cos x$
5. $y - x \frac{dy}{dx} = \frac{dy}{dx} y^2 e^y$
6. $(x^3 + xy^4) \, dx + 2y^3 \, dy = 0$
7. $\frac{dy}{dx} - y = y^2 e^x$

Exact Differential Equation

1. $(x - 2e^y) \, dy + (y + x \sin x) \, dx = 0$
2. $(x^2 + y^2) \, dx - 2xy \, dy = 0$

3. Is the differential equation $\left(1 + e^{\frac{x}{y}}\right) dx + e^{\frac{x}{y}} \left(1 - \frac{x}{y}\right) dy = 0$ exact? Solve this equation.
(Au-23)
4. $2x \sin y \, dx + x^2 \cos y \, dy = 0$
5. $e^y \, dx + (xe^y + 2y) \, dy = 0$
6. $(9x^2 + y - 1) - (4y - x) \frac{dy}{dx} = 0$
7. $(2x y^2 + 2y) + (2x^2 y + 2x) \frac{dy}{dx} = 0$
8. $(x^2 - y^2) \, dx - 2xy \, dy = 0$

Higher Order Differential Equation with Constants Co-efficients

1. $\frac{d^3 y}{dx^3} - 13 \frac{dy}{dx} - 12y = 0$
2. $(D^3 - 2D^2 - 4D + 8) y = 0$
3. $\frac{d^4 y}{dx^4} - a^4 y = 0$
4. $(D^2 + 5D + 6) y = 0$
5. $(D^2 - 6D + 9)y = 0$
6. $(D^2 - 2D + 5) y = 0$
7. $(D^4 - 2D^3 + 5D^2 - 8D + 4) y = 0$
8. $(D^2 - 3D + 2) y = 0, x = 0, y = 0, Dy = 0$
9. $(D^4 - 2D^3 + 5D^2) y = 0$
10. $(D^2 - 3D + 2) y = 1$
11. $(D^3 - 3D^2 + 4D - 2) y = e^{-2x} - \cos x$
12. $(D^3 - 3D^2 + 4D - 2) y = e^{-x} - \sin 2x$
13. $(D^3 - D) y = e^{-x} + \cos 2x$. (Au-23)
14. $(D^2 + 4D + 4) y = e^{-2x} + \sin 2x$
15. $(D^3 + 3D^2 + 3D + 1) y = e^{-x}$

International Islamic University Chittagong Department Of Computer Science and Engineering Differential Equation Group- B

Method of Variation of Parameter

1. $(D^2 - 1) y = \frac{2}{1 + e^x}$
2. $(D^2 + 1) y = \operatorname{cosec} x$ (Au-23)
3. $\frac{d^2 y}{dx^2} + n^2 y = \sec nx$
4. $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2 e^x$
5. $(D^2 - 1) y = e^{-x} \sin e^{-x} + \cos e^{-x}$

6. $(D^2 + 4)y = 4 \tan 2x$
7. $(1 - x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = (1 - x)^2$
8. $x^2 \frac{d^2 y}{dx^2} - 2x(1 + x) \frac{dy}{dx} + 2(1 + x)y = x^3$

Method of Undetermined Co-efficients

1. $(D^2 + 2)y = e^x + 2$
2. $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} - 5y = x^2 + 2e^{3x}$
3. $(D^2 - 2D + 3)y = x^3 + \sin x$ (Au-23)
4. $(D^2 - 2D)y = e^x \sin x$
5. $(D^3 + 2D^2 - D + 2)y = e^x + x^2$
6. $(D^2 + 2D + 2)y = x^2 + \sin x$
7. $(D^3 + 3D^2 + 2D)y = x^2 + 4x + 8$

Bessel's Equation

1. Define Bessel's function. Derive the solution of Bessel's equation.
2. i. Prove that, $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$
 ii. Prove that, $J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$
3. i. Show that, $\sqrt{\frac{\pi x}{2}} J_{\frac{3}{2}}(x) = \frac{\sin x}{x} - \cos x$
 ii. Show that, $\sqrt{\frac{\pi x}{2}} J_{-\frac{3}{2}}(x) = \frac{\cos x}{x} - \sin x$
4. Show that, $\int_0^x x^{n+1} J_n(x) dx = x^{n+1} J_{n+1}(x)$
5. Prove that, $J_n(-x) = (-1)^n J_n(x)$ for +ve or -ve integers.

Legendre's Equation

1. Define Legendre's equation. Derive the Legendre's polynomial $P_n(x)$ and $Q_n(x)$.
2. Show that, i. $P_n(1) = 1$
 ii. $P_n(-1) = (-1)^n$

Rodrigue's Formula

1. Define Rodrigue's formula. Using the Rodrigue's formula evaluate the value of $P_2(x), P_1(x), P_0(x)$. (Au-23)
2. State and prove the orthogonal condition of Legendre polynomial.
 or, Prove that, $\int_{-1}^1 P_m(x) P_n(x) dx = 0$ if $n \neq m$.
3. Show that, $\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}$
4. Show that, $\int_{-1}^1 [P_n(x)] dx = 0$, when $n \neq 0$

$$= 2, \text{ when } n = 0$$

Partial Differential Equation

Linear Partial Differential Equation of Order one

1. Define Linear Partial Differential Equation of Order one with example.
2. Find the differential equation from, $\phi(x + y + z, x^2 + y^2 - z^2) = 0$.
3. Form the differential equation whose solution is given by, $y = f(x - at) + F(x + at)$

Lagrange's Method

1. Solve: $yz p + xz q = xy$. (Au-23)
2. Solve: $xz p + yz q = z^2$
3. $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$
4. $(y + z)p + (z + x)q = x + y$
5. $(z^2 - 2yz - y^2)p + (xy + xz)q = xy - xz$
6. $x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$
7. $(y^2 + z^2 - x^2)p - 2xyq + 2xz = 0$
8. $(y^3x - 2x^4)p + (2y^4 - x^3y)q = 9z(x^3 - y^3)$
9. $pz - qz = z^2 + (x + y)^2$
10. $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$

Charpit's Method

1. Find the complete integral of the equation, $px + qy = pq$
2. Solve: $(p^2 + q^2)y = qz$.
3. $2xz - px^2 - 2qxy + pq = 0$. (Au-23)
4. $q = -xp + p^2$
5. $p = (qy + z)^2$
6. $pxy + pq + qy - yz = 0$

Higher Order Linear Partial Differential Equation

1. Solve : $(D^3 - 3D^2D' + 2D'^2D)z = 0$
2. $(D^4 - D'^4)z = 0$
3. $(D^3 - 6D^2D' + 11DD'^2 - 6D'^3)z = 0$
4. $(D^4 - 2D^3D' + 2D^2D'^2 - D'^4)z = 0$
5. $r - 4s + 4t = 0$ or, $(D^2 - 4DD' + 4D'^2)z = 0$
6. $(D^3 - 3D^2D' + 3DD'^2 - D'^3)z = 0$
7. $(D^3 - 2D^2D' + DD'^2)z = 0$
8. $(4D^2 + 12DD' + 9D'^2)z = 0$
9. $(D^2 - 6DD' + 9D'^2)z = 12x^2 + 36xy$
10. $(D^2 - D'^2)z = x - y$
11. $(D^2 - a^2D'^2)z = x$
12. $(D^2 - DD' - 6D'^2)z = xy$
13. $(D^3 - 7DD'^3 - 6D'^3)z = x^2 + xy^2 + y^3$

A short Method

1. $(D^2 + 2DD' + D'^2)z = e^{2x+3y}$
2. $(D^2 + 3DD' + 2D'^2)z = x + y$
3. $(D^2 + D'^2)z = 30(2x + y)$
4. $(2D^2 - 5DD' + 2D'^2)z = 24(y - x)$
5. $(D^2 - 2DD' + D'^2)z = e^{x+2y}$
6. $(D + D')z = \sin x$
7. $(D^3 - 4D^2D' + 4DD'^2)z = 4 \sin(2x + y)$
8. $(D^3 - 2D^2D' - DD'^2 + 2D'^3)z = e^{x+y}$
9. $(2D^2 - DD' - 3D'^2)z = 5e^{x+y}$
10. $(D^2 - 5DD' + 4D'^2)z = \sin(4x + y)$

Application of Differential Equation

1. Bacteria in a certain culture increase at a rate proportional to the number present. If the number doubles in one hour, how long does it take for the number to triple?
2. When a cake is removed from an oven, its temperature is measured at 300°F. Three minutes later its temperature is 200°F. How long will it take to cool off to a room temperature of 70°F?
3. The body of a murder victim was discovered at 11:00 p.m. The doctor took the temperature of the body at 11:30 p.m. which was 94.6°F. He again took the temperature after one hour when showed 93.4°F and noticed that the temperature of the room was 70°F. Estimate the time of death. (Normal temperature of human body is 98.4°F). (Au-23)
4. A generator having emf 100V is connected in series with a 10Ω resistor and an inductor of 2H. If the switch K is closed at time $t = 0$. Obtain a differential equation for the current and determine the current at time t . (Au-23)